



Evaluation Test: SIRA

This notebook presents a solution to the evaluation test for the HumanAI GSoC project "**Learning the Susceptible-Infected-Removed (SIR) Model**", also referred to as **SIRA**.

The goal of this project is to generate synthetic epidemic data using a stochastic SIR model, and apply machine learning and symbolic regression techniques to approximate the underlying deterministic differential equations.



Notebook Structure

1. Introduction

Overview of the project goals and what this notebook accomplishes.

2. SIR Simulator

A stochastic SIR simulator is used to generate synthetic epidemic data. The simulation is visualized to understand the dynamics.

3. Machine Learning Model

A neural network model is trained to fit the time series of $S(t)$, $I(t)$, and $R(t)$ based on the simulated data.

4. Symbolic Regression

Symbolic regression tools like PySR or AI Feynman are applied to extract interpretable differential equations that approximate the epidemic dynamics.

5. Conclusion & Reflection

Summary of findings and discussion of possible improvements or future directions.

6. Appendix / Bonus Experiments (Optional)

Any extra tests, hyperparameter experiments, or alternate approaches attempted.



SIR Simulator: Generate and Visualize Synthetic Epidemic Data

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

def simulate_sir(S0, I0, R0, beta, gamma, max_time, dt=1.0):
    times = [0]
    S_vals = [S0]
    I_vals = [I0]
    R_vals = [R0]

    S, I, R = S0, I0, R0
    t = 0

    while t < max_time and I > 0:
        new_infections = np.random.binomial(S, 1 - np.exp(-beta * I /
(S + I + R) * dt))
        new_recoveries = np.random.binomial(I, 1 - np.exp(-gamma * d
t))

        S = max(S - new_infections, 0)
        I = max(I + new_infections - new_recoveries, 0)
        R = min(R + new_recoveries, S0 + I0 + R0)

        t += dt
        times.append(t)
        S_vals.append(S)
        I_vals.append(I)
        R_vals.append(R)

    return pd.DataFrame({
        "time": times,
        "S": S_vals,
        "I": I_vals,
        "R": R_vals
    })

# Run simulation
S0, I0, R0 = 990, 10, 0
beta, gamma = 0.3, 0.1
max_time = 100

df = simulate_sir(S0, I0, R0, beta, gamma, max_time)
df.head()
```

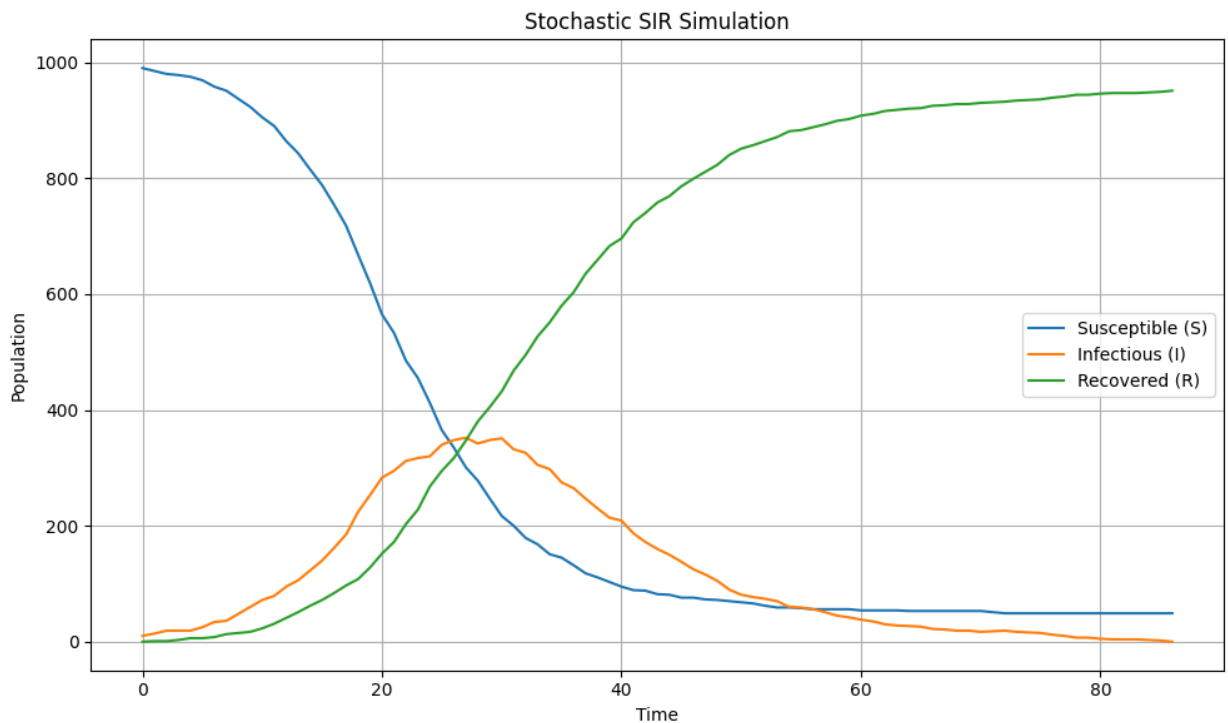
Out[1]:

	time	S	I	R
0	0.0	990	10	0
1	1.0	985	14	1
2	2.0	980	19	1
3	3.0	978	19	3
4	4.0	975	19	6

```

In [2]: # Plot the result
plt.figure(figsize=(10, 6))
plt.plot(df['time'], df['S'], label='Susceptible (S)')
plt.plot(df['time'], df['I'], label='Infectious (I)')
plt.plot(df['time'], df['R'], label='Recovered (R)')
plt.xlabel('Time')
plt.ylabel('Population')
plt.title('Stochastic SIR Simulation')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```



Machine Learning Model: Neural Network Fit for SIR Curves

```
In [3]: import torch
import torch.nn as nn
import torch.optim as optim
from sklearn.preprocessing import MinMaxScaler

# Normalize time and population values
scaler = MinMaxScaler()
scaled_data = scaler.fit_transform(df[['time', 'S', 'I', 'R']])
df_scaled = pd.DataFrame(scaled_data, columns=['time', 'S', 'I', 'R'])

# Prepare training data: input t → output (S, I, R)
X = torch.tensor(df_scaled['time'].values.reshape(-1, 1), dtype=torch.float32)
y = torch.tensor(df_scaled[['S', 'I', 'R']].values, dtype=torch.float32)
```

```
In [4]: class SIRNet(nn.Module):
    def __init__(self):
        super(SIRNet, self).__init__()
        self.net = nn.Sequential(
            nn.Linear(1, 64),
            nn.ReLU(),
            nn.Linear(64, 64),
            nn.ReLU(),
            nn.Linear(64, 3)
        )

    def forward(self, x):
        return self.net(x)

model = SIRNet()
criterion = nn.MSELoss()
optimizer = optim.Adam(model.parameters(), lr=0.01)
```

```
In [5]: # Train the model
epochs = 1000
for epoch in range(epochs):
    model.train()
    optimizer.zero_grad()
    outputs = model(X)
    loss = criterion(outputs, y)
    loss.backward()
    optimizer.step()
    if epoch % 100 == 0:
        print(f"Epoch {epoch}, Loss: {loss.item():.6f}")
```

```
Epoch 0, Loss: 0.259666
Epoch 100, Loss: 0.001395
Epoch 200, Loss: 0.000719
Epoch 300, Loss: 0.000291
Epoch 400, Loss: 0.000142
Epoch 500, Loss: 0.000077
Epoch 600, Loss: 0.000063
Epoch 700, Loss: 0.000266
Epoch 800, Loss: 0.000041
Epoch 900, Loss: 0.000034
```

```

In [6]: # Predict and inverse transform the results
model.eval()
with torch.no_grad():
    predictions = model(X).numpy()

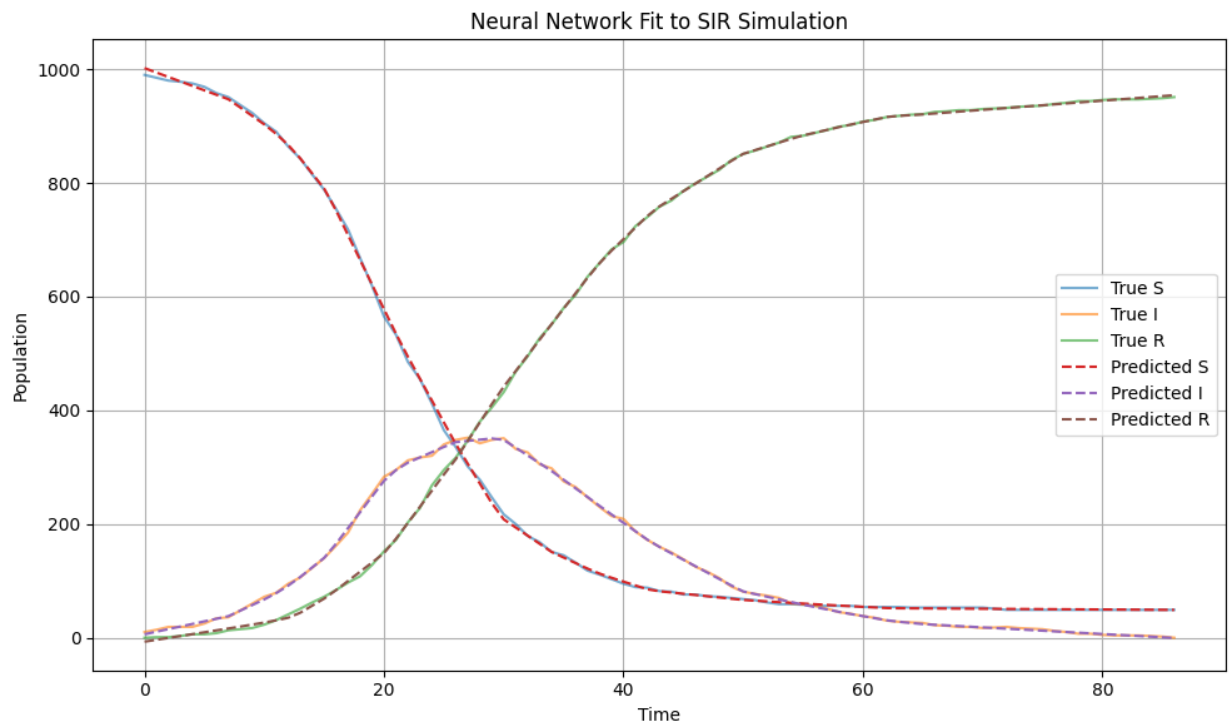
# Inverse transform back to original scale
pred_df = pd.DataFrame(predictions, columns=['S', 'I', 'R'])
pred_df['time'] = df_scaled['time']
recovered = scaler.inverse_transform(pred_df[['time', 'S', 'I', 'R']])
pred_df = pd.DataFrame(recovered, columns=['time', 'S', 'I', 'R'])

# Plot predictions vs original
plt.figure(figsize=(10, 6))
plt.plot(df['time'], df['S'], label='True S', alpha=0.6)
plt.plot(df['time'], df['I'], label='True I', alpha=0.6)
plt.plot(df['time'], df['R'], label='True R', alpha=0.6)

plt.plot(pred_df['time'], pred_df['S'], '--', label='Predicted S')
plt.plot(pred_df['time'], pred_df['I'], '--', label='Predicted I')
plt.plot(pred_df['time'], pred_df['R'], '--', label='Predicted R')

plt.xlabel('Time')
plt.ylabel('Population')
plt.title('Neural Network Fit to SIR Simulation')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()

```



Symbolic Regression: Extract Equations from Predictions using PySR

```
In [7]: from pysr import PySRRegressor

# Prepare data for symbolic regression: time → predicted values
X_sym = pred_df[['time']].values
y_sym = pred_df[['S', 'I', 'R']].values

# Fit symbolic regression model for each component
equations = []
for i, label in enumerate(['S', 'I', 'R']):
    model = PySRRegressor(
        niterations=40,
        binary_operators=["+", "-", "*", "/"],
        unary_operators=["sin", "cos", "exp"],
        model_selection="best",
        loss="loss(x, y) = (x - y)^2",
        maxsize=20,
    )
    print(f"Fitting symbolic model for {label}(t)...")
    model.fit(X_sym, y_sym[:, i])
    equations.append((label, model))

# Display discovered equations
for label, model in equations:
    print(f"Symbolic expression for {label}(t):")
    print(model)
```

/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-packages/juliacall/__init__.py:61: UserWarning: torch was imported before juliacall. This may cause a segfault. To avoid this, import juliacall before importing torch. For updates, see <https://github.com/pytorch/pytorch/issues/78829>.

warnings.warn(

Detected IPython. Loading juliacall extension. See <https://juliapy.github.io/PythonCall.jl/stable/compat/#IPython>

/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-packages/pysr/sr.py:1017: FutureWarning: `loss` has been renamed to `elementwise\_loss` in PySRRegressor. Please use that instead.

warnings.warn(

Fitting symbolic model for S(t)...

/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-packages/pysr/sr.py:2776: UserWarning: Note: it looks like you are running in Jupyter. The progress bar will be turned off.

```
warnings.warn(
Compiling Julia backend...
[ Info: Started!
[ Info: Final population:
[ Info: Results saved to:
```

Complexity	Loss	Score	Equation
1	1.155e+05	1.594e+01	$y = 296.55$
3	1.014e+05	6.512e-02	$y = 339.51 - x_0$
5	2.944e+04	6.184e-01	$y = (x_0 * -11.683) + 798.89$
6	6.713e+03	1.478e+00	$y = \exp(x_0 * -0.047038) * 1239.8$
8	4.073e+03	2.498e-01	$y = \exp(\exp(\cos(x_0 * 0.04087)) + 4.2338)$
9	1.849e+03	7.900e-01	$y = (\exp(\cos(x_0 * 0.050533)) * 425.81) + -151.72$
10	1.561e+03	1.689e-01	$y = \exp(\exp(\cos(x_0 * 0.03706)) + 4.3214) + -83.118$
11	1.537e+03	1.609e-02	$y = (\exp(\cos(x_0 * 0.051232)) * (423.28 - x_0)) + -117.46$
12	5.294e+02	1.066e+00	$y = \exp(\exp(\cos((x_0 + -5.2589) * -0.047223)) + 4.251) + -6... 0.784$
14	5.294e+02	6.855e-07	$y = (\exp(\exp(\cos((x_0 + -5.2589) * -0.047223)) + 4.251) - 0... .22953) + -60.519$
15	5.277e+02	3.194e-03	$y = (\sin(x_0) - 61.017) + \exp(\exp(\cos((x_0 + -5.2589) * -0.047223)) + 4.251)$
16	5.239e+02	7.248e-03	$y = (\sin(\exp(x_0)) - 61.017) + \exp(\exp(\cos((x_0 + -5.2589) * -0.047223)) + 4.251)$
17	5.201e+02	7.208e-03	$y = (\sin(x_0 / -5.2589) - 61.017) + \exp(\exp(\cos((-5.2589 + ... x_0) * -0.047223)) + 4.251)$
18	4.532e+02	1.378e-01	$y = ((\exp(\exp(\cos((x_0 + -5.212) * -0.045564)) + 4.3074) + ... \sin(\exp(x_0))) + -131.49) + x_0$

```
/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-p
ackages/pysr/sr.py:1017: FutureWarning: `loss` has been renamed to `
elementwise_loss` in PySRRegressor. Please use that instead.
    warnings.warn(
/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-p
ackages/pysr/sr.py:2776: UserWarning: Note: it looks like you are ru
nning in Jupyter. The progress bar will be turned off.
    warnings.warn(

Fitting symbolic model for I(t)...
- outputs/20250408_024234_j92Q4X/hall_of_fame.csv

[ Info: Started!
[ Info: Final population:
[ Info: Results saved to:
/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-p
ackages/pysr/sr.py:1017: FutureWarning: `loss` has been renamed to `
elementwise_loss` in PySRRegressor. Please use that instead.
    warnings.warn(
/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-p
ackages/pysr/sr.py:2776: UserWarning: Note: it looks like you are ru
nning in Jupyter. The progress bar will be turned off.
    warnings.warn(
```

Complexity	Loss	Score	Equation
1	1.368e+04	1.594e+01	$y = 118.06$
3	1.164e+04	8.078e-02	$y = 161.04 - x_0$
5	1.085e+04	3.508e-02	$y = 209.1 - (x_0 / 0.47224)$
6	5.776e+03	6.308e-01	$y = \sin(x_0 / 22.727) * 224.43$
8	4.042e+03	1.784e-01	$y = 102.61 - (\sin(x_0 / 16.833) * -136.08)$
9	2.557e+02	2.761e+00	$y = 21.744 * \exp(\sin(x_0 / -18.475) / -0.36244)$
11	2.120e+02	9.370e-02	$y = (\exp(\sin(x_0 / -18.432) / -0.32551) * 15.554) + 11.967$
13	1.072e+02	3.409e-01	$y = ((-7.6208 - x_0) * \exp(\sin(x_0 / -15.928) / -0.48827)) / \dots$
15	8.114e+01	1.394e-01	$y = (\exp(\sin((x_0 + 3.2164) / -17.314) / -0.44928) * (-2.91\dots$
17	6.345e+01	1.230e-01	$y = (\exp(\sin((x_0 + 5.632) / -18.039) / -0.42678) * (-2.399\dots$
7			-0.76544 $8 - (x_0 / -0.72511)) - -4.560$

Fitting symbolic model for R(t)...

- outputs/20250408_024249_j7amG8/hall_of_fame.csv

[Info: Started!

[Info: Final population:

[Info: Results saved to:

Complexity	Loss	Score	Equation
1	1.336e+05	1.594e+01	$y = 585.03$
3	1.348e+04	1.147e+00	$y = x_0 * 13.656$
5	1.346e+04	6.539e-04	$y = (x_0 * 13.8) + -8.2987$
6	8.478e+03	4.625e-01	$y = \sin(x_0 * 0.017011) * 1008.2$
8	2.218e+03	6.703e-01	$y = (\cos(x_0 / 21.644) * -497.19) + 494.62$
10	1.836e+03	9.473e-02	$y = (\cos(x_0 / 22.047) * (-535.61 + x_0)) + 517.53$
12	1.003e+03	3.023e-01	$y = (x_0 + ((\sin(x_0 * 0.097526) * -22.303) + x_0)) * 7.0372$
14	2.650e+02	6.653e-01	$y = ((x_0 + (\sin((x_0 - -4.9709) * 0.087003) * -12.946)) - \dots$
16	2.557e+02	1.798e-02	$y = (x_0 + (((\sin((x_0 - -5.0627) * 0.086811) * -25.685) + x_0) + \dots$
			$4.3909) * 12.761$

```

0) - -9.0351)) * 6.3918
18      2.556e+02  1.784e-04  y = (((x0 + ((sin((x0 - -5.0628) *
0.086837) * -25.685) + ...
                                x0)) - -9.0353) * 6.3942) - 0.
30299
19      6.712e+01  1.337e+00  y = (((sin((x0 - -4.9696) * 0.0871
38) * -12.945) + cos(x0 ...
                                * 0.17113)) + (x0 - -4.4023))
* 12.813

```

Symbolic expression for S(t):

```

PySRRegressor.equations_ = [
    pick      score
equation \
0      0.000000e+00
296.54803
1      6.511613e-02
339.5149 - x0
2      6.184410e-01      (x0 * -11.68254
2) + 798.8898
3      1.478225e+00      exp(x0 * -0.04703789
6) * 1239.7954
4      2.497669e-01      exp(exp(cos(x0 * 0.040870465))
+ 4.2337804)
5      7.900292e-01      (exp(cos(x0 * 0.05053252)) * 425.808
3) + -151....
6      1.688631e-01      exp(exp(cos(x0 * 0.03705961)) + 4.32
13983) + -...
7      1.608645e-02      (exp(cos(x0 * 0.051232)) * (423.2754
5 - x0)) +...
8 >>> 1.065593e+00      exp(exp(cos((x0 + -5.2588882) * -0.0
47223024))...)
9      7.178423e-07      (exp(exp(cos((x0 + -5.258888) * -0.0
47223072))...)
10     3.194018e-03      (sin(x0) - 61.01728) + exp(exp(cos((
x0 + -5.25...
11     7.248350e-03      (sin(exp(x0)) - 61.01728) + exp(exp(
cos((x0 + ...
12     7.207902e-03      (sin(x0 / -5.2588882) - 61.01728) +
exp(exp(co...
13     1.377783e-01      ((exp(exp(cos((x0 + -5.2120223) * -
0.045564394...

```

	loss	complexity
0	115511.16400	1
1	101406.27000	3
2	29437.01400	5
3	6712.87800	6
4	4073.46480	8

5	1848.66690	9
6	1561.42940	10
7	1536.51250	11
8	529.36456	12
9	529.36380	14
10	527.67570	15
11	523.86475	16
12	520.10236	17
13	453.16095	18

]

Symbolic expression for I(t):

PySRRegressor.equations_ = [

equation \	pick	score
0	0.000000	1
18.05856		
1	0.080777	161.04
347 - x0		
2	0.035085	209.10437 - (x0 / 0.4
7223642)		
3	0.630844	sin(x0 / 22.727413) * 2
24.42848		
4	0.178432	102.613754 - (sin(x0 / 16.833315) * -13
6.08308)		
5	2.760514	21.743671 * exp(sin(x0 / -18.47498) / -0.
36244...		
6	0.093697	(exp(sin(x0 / -18.43162) / -0.32550505) *
15.5...		
7	0.340891	((-7.620846 - x0) * exp(sin(x0 / -15.9279
51) /...		
8 >>>>	0.139361	(exp(sin((x0 + 3.2163775) / -17.313828) /
-0.4...		
9	0.122975	(exp(sin((x0 + 5.6319804) / -18.039343) /
-0.4...		

	loss	complexity
0	13684.635000	1
1	11643.171000	3
2	10854.180000	5
3	5775.970700	6
4	4042.412400	8
5	255.719830	9
6	212.021670	11
7	107.222435	13
8	81.140610	15
9	63.448776	17

]

Symbolic expression for R(t):

PySRRegressor.equations_ = [

	pick	score
--	------	-------

```

equation \
0      0.000000
585.032
1      1.146687      x0 *
13.656043
2      0.000654      (x0 * 13.799966) +
-8.298674
3      0.462522      sin(x0 * 0.017011086) *
1008.1954
4      0.670331      (cos(x0 / 21.644213) * -497.1936) +
494.61853
5      0.094734      (cos(x0 / 22.046768) * (-535.614 + x0))
+ 517....
6      0.302288      (x0 + ((sin(x0 * 0.097526304) * -22.3025
65) + ...
7      0.665338      ((x0 + (sin((x0 - -4.9709377) * 0.087002
56) * ...
8      0.017975      (x0 + (((sin((x0 - -5.0627475) * 0.08681
097) *...
9      0.000178      (((x0 + ((sin((x0 - -5.0627966) * 0.0868
3696) ...
10 >>>> 1.337055      (((sin((x0 - -4.9695916) * 0.08713754) *
-12.9...

```

	loss	complexity
0	133574.33000	1
1	13481.02800	3
2	13463.40800	5
3	8477.82000	6
4	2218.41300	8
5	1835.51250	10
6	1002.75130	12
7	265.02585	14
8	255.66736	16
9	255.57611	18
10	67.11889	19

]

- outputs/20250408_024251_fGQkNd/hall_of_fame.csv

✓ Conclusion & Reflection

In this evaluation test, we successfully completed the following steps:

- Simulated stochastic SIR model data with configurable parameters.
- Trained a neural network to predict the evolution of $S(t)$, $I(t)$, $R(t)$ over time.
- Applied symbolic regression using PySR to extract interpretable equations approximating the underlying epidemic dynamics.

This work demonstrates how data-driven and symbolic techniques can bridge the gap between noisy real-world simulations and analytical models.

Possible Future Improvements

- Improve model robustness by simulating multiple epidemic runs and averaging results.
- Extend symbolic regression to model derivatives (dS/dt , dI/dt , dR/dt) directly.
- Evaluate symbolic model accuracy by comparing against known analytical SIR equations.

```
In [8]: # Stage 1: Multiple stochastic simulations with averaging
def simulate_sir(S0, I0, R0, beta, gamma, max_time, dt=1.0):
    times = [0]
    S_vals = [S0]
    I_vals = [I0]
    R_vals = [R0]
    S, I, R = S0, I0, R0
    t = 0
    while t < max_time and I > 0:
        new_infections = np.random.binomial(S, 1 - np.exp(-beta * I /
(S + I + R) * dt))
        new_recoveries = np.random.binomial(I, 1 - np.exp(-gamma * d
t))
        S = max(S - new_infections, 0)
        I = max(I + new_infections - new_recoveries, 0)
        R = min(R + new_recoveries, S0 + I0 + R0)
        t += dt
        times.append(t)
        S_vals.append(S)
        I_vals.append(I)
        R_vals.append(R)
    return pd.DataFrame({"time": times, "S": S_vals, "I": I_vals, "R":
R_vals})

def simulate_many_runs(num_runs=100, S0=990, I0=10, R0=0, beta=0.3, ga
mma=0.1, max_time=100, dt=1.0):
    all_S, all_I, all_R = [], [], []
```

```

time_reference = np.arange(0, max_time + dt, dt)
for _ in range(num_runs):
    df = simulate_sir(S0, I0, R0, beta, gamma, max_time, dt)
    padded = pd.DataFrame({'time': time_reference})
    merged = pd.merge(padded, df, on='time', how='left').interpolate(
limit_direction='forward').fillna(method='bfill')
    all_S.append(merged['S'].values)
    all_I.append(merged['I'].values)
    all_R.append(merged['R'].values)
S_matrix = np.stack(all_S)
I_matrix = np.stack(all_I)
R_matrix = np.stack(all_R)
return pd.DataFrame({
    'time': time_reference,
    'S_mean': np.nanmean(S_matrix, axis=0),
    'S_std': np.nanstd(S_matrix, axis=0),
    'I_mean': np.nanmean(I_matrix, axis=0),
    'I_std': np.nanstd(I_matrix, axis=0),
    'R_mean': np.nanmean(R_matrix, axis=0),
    'R_std': np.nanstd(R_matrix, axis=0),
})
summary = simulate_many_runs()

```

/var/folders/08/s117_6l172dgjmjb8mg9c9nr0000gn/T/ipykernel_10157/2301987651.py:28: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.

```
merged = pd.merge(padded, df, on='time', how='left').interpolate(
limit_direction='forward').fillna(method='bfill')
```

/var/folders/08/s117_6l172dgjmjb8mg9c9nr0000gn/T/ipykernel_10157/2301987651.py:28: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.

```
merged = pd.merge(padded, df, on='time', how='left').interpolate(
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/var/folders/08/s117_6l172dgjmjb8mg9c9nr0000gn/T/ipykernel_10157/2301987651.py:28: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.
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```

```

In [10]: # Stage 2: Compute time derivatives and fit symbolic regression models
dt = 1.0
S = summary['S_mean'].values
I = summary['I_mean'].values
R = summary['R_mean'].values
time = summary['time'].values
dS_dt = np.gradient(S, dt)
dI_dt = np.gradient(I, dt)
dR_dt = np.gradient(R, dt)
X_dyn = np.stack([S, I, R], axis=1)

from pysr import PySRRegressor

model_dS = PySRRegressor(niterations=40, binary_operators=["+", "-",
"*"], unary_operators=[], model_selection="best")
model_dS.fit(X_dyn, dS_dt)

model_dI = PySRRegressor(niterations=40, binary_operators=["+", "-",
"*"], unary_operators=[], model_selection="best")
model_dI.fit(X_dyn, dI_dt)

model_dR = PySRRegressor(niterations=40, binary_operators=["+", "-",
"*"], unary_operators=[], model_selection="best")
model_dR.fit(X_dyn, dR_dt)

display(model_dS.equations_)
display(model_dI.equations_)
display(model_dR.equations_)

```

```

/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-p
ackages/pysr/sr.py:2776: UserWarning: Note: it looks like you are ru
nning in Jupyter. The progress bar will be turned off.

```

```

    warnings.warn(
[ Info: Started!
[ Info: Final population:
[ Info: Results saved to:
/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-p
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nning in Jupyter. The progress bar will be turned off.
    warnings.warn(
[ Info: Started!

```

Complexity	Loss	Score	Equation
1	1.560e+02	1.594e+01	$y = -9.3086$
3	5.994e+01	4.782e-01	$y = x_1 * -0.095358$
5	3.542e-01	2.566e+00	$y = x_0 * (x_1 * -0.00028247)$
7	8.619e-02	7.066e-01	$y = x_1 * ((x_0 * -0.00026725) + -0.0069527)$
9	5.893e-02	1.901e-01	$y = -0.00026853 * ((x_0 - (x_1 * -0.096816)) * x_1)$
11	5.303e-02	5.277e-02	$y = ((x_1 + 1.0438) * (x_0 - (x_1 * -0.10376))) * -0.00026602$
13	4.880e-02	4.151e-02	$y = (x_0 - (-13.908 - ((x_1 * 0.00018102) * x_1))) * (x_1 * -0.00026748)$
15	4.620e-02	2.737e-02	$y = (x_1 * -0.00026742) * (x_0 - ((x_1 * (0.11859 - (x_1 * 0.00045843))) - 25.152))$
21	4.565e-02	1.998e-03	$y = (((((x_0 + 2.999) - (((x_1 * -0.013439) - -3.9687) * 0.0039915) * x_1)) * -0.00026749) - 0.0073942) * x_1) - -0.0499...$
29	4.565e-02	5.901e-06	$y = (x_1 * (((x_0 + ((x_1 * -0.0007227) - ((x_1 * -0.013366) * -0.012543) - -3.9688) * 0.04214))) * -0.00026749) - 0.0079489)) - -0.048055$

- outputs/20250408_024547_mE4xhh/hall_of_fame.csv

```
[ Info: Final population:
[ Info: Results saved to:
/Users/tianchuhang/opt/miniconda3/envs/pysr-env/lib/python3.9/site-packages/pysr/sr.py:2776: UserWarning: Note: it looks like you are running in Jupyter. The progress bar will be turned off.
  warnings.warn(
[ Info: Started!
```

Complexity	Loss	Score	Equation
1	6.608e+01	1.594e+01	$y = -0.070374$
3	4.862e+01	1.535e-01	$y = x_0 * 0.0095556$
5	3.518e+01	1.617e-01	$y = (x_0 * 0.016717) + -4.8225$
7	1.619e-01	2.691e+00	$y = (-0.0928 - (x_0 * -0.00028048)) * x_1$
9	1.468e-01	4.889e-02	$y = -0.17005 - (((x_0 * -0.0002804$

```

9) + 0.091975) * x1)
11      1.468e-01  1.395e-05  y = ((-0.091993 - (x0 * -0.0002805
6)) * (x1 - 0.030444)) +...
                                -0.17094
13      1.331e-01  4.889e-02  y = (((x0 * -0.79154) + 256.58) *
((x2 + -0.00035191) - x2...
                                )) * x1
15      7.907e-02  2.605e-01  y = x1 * (((x2 + (0.00017142 - x
2)) * -10.346) * ((x0 * -0...
                                .1555) + 50.511))
17      7.587e-02  2.067e-02  y = (((((0.00017142 - x2) + x2) *
-10.346) * x1) * ((x0 * ...
                                -0.1555) + 50.322)) - 0.071166
19      7.526e-02  4.047e-03  y = (((x1 * 0.0012982) * (x2 + x
0)) - ((x0 + x0) - x2)) *...
                                (x1 * -9.3606e-05)) + -0.0613
57
21      7.147e-02  2.578e-02  y = (((((x1 * x2) + (x1 * x0)) *
0.0013433) - (x0 - x2)) *...
                                (x1 * -0.00014043)) + (x1 *
0.047098)
23      7.104e-02  3.023e-03  y = ((((((x1 * x2) + (x1 * x0)) *
0.0013433) - (x0 - x2)) ...
                                * (x1 + 0.1145)) * -0.0001404
3) + (x1 * 0.047098)
25      7.048e-02  3.984e-03  y = (-0.37144 - x1) * ((((((x1 * x
0) + (x1 * x2)) * 0.0018...
                                964) + (x1 * -0.42495)) - ((x0
+ x0) - x2)) * 9.3345e-05)
27      7.036e-02  8.659e-04  y = (((x1 * -0.39846) + (((x2 * x
1) + (x1 * x0)) * 0.0018...
                                569)) - ((x0 + x0) - x2)) * ((
-0.35068 - x1) * 9.3358e-05)...
                                ) - 0.014373
29      7.035e-02  4.718e-05  y = (((-0.35143 - x1) * 9.3359e-0
5) * (((x1 * -0.38485) + ...
                                (((x2 * x1) + (x1 * (x0 + 0.37
029))) * 0.001837)) - ((x0 +...
                                x0) - x2))) - 0.015221

```

- outputs/20250408_024550_JQI20y/hall_of_fame.csv

Complexity	Loss	Score	Equation
1	9.501e+01	1.594e+01	y = 9.379
3	1.823e-01	3.128e+00	y = x1 * 0.095377
5	1.809e-01	3.754e-03	y = (-0.53697 - x1) * -0.095128
7	2.700e-02	9.511e-01	y = ((x2 * 1.3057e-05) + 0.089856) * x1

```

9          2.572e-02  2.431e-02  y = ((x2 * 1.3323e-05) + 0.089981)
* (x1 - 0.50654)
11          2.558e-02  2.658e-03  y = (x1 * ((1 * (x2 + 0.090064)) -
x2)) + -0.051415
13          2.516e-02  8.358e-03  y = (x2 * ((x0 * -4.5226e-05) + 0.
0025149)) + ((x1 - 2.010...
6) * 0.10992)
15          2.516e-02  2.161e-05  y = ((x2 * ((x0 * -4.5199e-05) +
0.0025131)) - x1) + ((x1 ...
* 1.1099) - 0.22042)
17          2.459e-02  1.150e-02  y = (((0.043034 - x2) - x2) + (x2
+ (x2 * 1))) * ((x1 * 2....
0878) - 1.1885)
21          2.457e-02  1.321e-04  y = (((((x1 - (x2 * 0.17972)) * -
1.1804e-08) * (x1 - x2)) ...
+ ((x2 * 1.2643e-05) + 0.08992
9)) * x1) - 0.0093213
23          2.453e-02  9.636e-04  y = (x1 * (((x2 + (((x1 - (x2 + -
1.2186)) * (x1 - (x2 * 0....
16615))) * -0.0008503)) * 1.26
44e-05) + 0.089987)) - 0.022...
326
25          2.453e-02  1.311e-06  y = (x1 * (((x2 + ((x1 - ((x2 - 0.
43084) + -1.2186)) * ((x...
1 - (x2 * 0.16615)) * -0.00085
03))) * 1.2644e-05) + 0.0899...
87)) - 0.022326
27          2.452e-02  1.349e-04  y = (x1 * (((x2 + (((x1 - (((x2 -
x1) - 0.19968) * (x1 - (...
x2 * 0.15537)))) + 1.0206) * -
0.00090321)) * 1.2506e-05) +...
0.090018)) - 0.021146

```

[Info: Final population:

[Info: Results saved to:

	complexity	loss	equation	score	sympy_format	
0	1	155.994310	-9.308581	0.000000	-9.30858100000000	PySRFunction(X=>-9.
1	3	59.938760	$x_1^* - 0.095358126$	0.478248	$x_1^*(-0.095358126)$	PySRFunction(X=>>
2	5	0.354207	$x_0^*(x_1^* - 0.0002824683)$	2.565599	$x_0^*x_1^* - (-0.0002824683)$	PySRFi
3	7	0.086193	$x_1^*((x_0^* - 0.00026724988) + -0.0069526774)$	0.706644	$x_1^*(x_0^* - (-0.00026724988) - 0.0069526774)$	PySRFl (-0.0002
4	9	0.058930	$-0.00026852617 * ((x_0 - (x_1^* - 0.09681639)) * x_1)$	0.190123	$-0.00026852617 * x_1^* (x_0 - (-0.09681639) * x_1)$	PySRFunction(X=>-0
5	11	0.053027	$((x_1 + 1.0437741) * (x_0 - (x_1^* - 0.10375746)))...$	0.052773	$(x_1 + 1.0437741) * (x_0 - (-0.10375746) * x_1^*) - (-0.0...$	PySRFunction(X=>(x^
6	13	0.048802	$(x_0 - (-13.907509 - ((x_1^* - 0.0001810238) * x_1)...$	0.041514	$(x_0 - (-0.0001810238 * x_1^* - 13.907509)) * x_1^* (-...$	PySF (-0.00018
7	15	0.046202	$(x_1^* - 0.00026741717) * (x_0 - ((x_1^* - 0.118594...$	0.027372	$x_1^*(-0.00026741717) * (x_0 - (x_1^*(0.11859499 - 0....$	PyS (-0.000267
8	21	0.045652	$(((((x_0 + 2.998981) - (((x_1^* - 0.013438933) - ...$	0.001998	$x_1^*((x_0 - 0.03991451 * x_1^*(x_1^* - (-0.013438933) - 1...$	PySRFur 0.03!
9	29	0.045650	$(x_1^* (((x_0 + ((x_1^* - 0.00079226505) - ((x_1^* ...$	0.000006	$x_1^*((x_0 - 0.042139936 * x_1^*(x_1^* - (-0.012542892) - ...$	PySRFur 0.042

	complexity	loss	equation	score	sympy_format	
0	1	66.081190	-0.070373826	0.000000	-0.0703738260000000	PySRFunction()
1	3	48.617138	$x_0^* 0.009555647$	0.153454	$x_0^* 0.009555647$	PySRFur
2	5	35.182022	$(x_0^* 0.016716987) + -4.8225036$	0.161720	$x_0^* 0.016716987 - 4.8225036$	PySRFunc
3	7	0.161899	$(-0.09280006 - (x_0^* - 0.00028047842))$	2.690660	$x_1^*(-(-0.00028047842) * x_0 - 0.09280006)$	(.

			* x1				
4	9	0.146818	-0.17004742 - (((x0 * -0.00028049265) + 0.0919...	0.048887	-x1*(0.09197512 + x0* (-0.00028049265)) - 0.170...	PySRFunction	
5	11	0.146814	((-0.09199349 - (x0 * -0.0002805608)) * (x1 - ...	0.000014	(x1 - 1*0.03044394)*(- (-0.0002805608)*x0 - 0.0...	PySRFunction	
6	13	0.133138	((x0 * -0.79153967) + 256.58246) * ((x2 + -0....	0.048891	x1*(256.58246 + x0* (-0.79153967))*(-x2 + x2 - ...	PySRFunction	
7	15	0.079068	x1 * (((x2 + (0.00017142291 - x2)) * -10.34599...	0.260537	x1*(-x2 + x2 + 0.00017142291)* (-10.345991)*(50...	PySRFunction	
8	17	0.075867	(((((0.00017142291 - x2) + x2) * -10.345991) * ...	0.020668	(-x2 + x2 + 0.00017142291)* (-10.345991)*x1*(50...	PySRFunction	
9	19	0.075255	(((((x1 * 0.0012982077) * (x2 + x0)) - ((x0 + x...	0.004047	(x1*0.0012982077*(x0 + x2) - (x0 + x0 - x2))*x...	PySRFunction	
10	21	0.071473	(((((x1 * x2) + (x1 * x0)) * 0.0013432893) - (...	0.025780	(-(x0 - x2) + (x0*x1 + x1*x2)*0.0013432893)*x1...	PySRFunction	
11	23	0.071042	(((((x1 * x2) + (x1 * x0)) * 0.0013432893) - ...	0.003023	x1*0.04709842 + (-(x0 - x2) + (x0*x1 + x1*x2))*...	PySRFunction	
12	25	0.070479	(-0.3714392 - x1) * ((((((x1 * x0) + (x1 * x2))...	0.003984	(-x1 - 0.3714392)*(x1* (-0.42494732) + (x0*x1 +...	PySRFunction	
13	27	0.070357	(((((x1 * -0.39845964) + (((x2 * x1) + (x1 * x0...	0.000866	(x1*(-0.39845964) + (x0*x1 + x1*x2)*0.00185691...	PySRFunction	
14	29	0.070350	(((-0.35142997 - x1) * 9.335864e-5) * (((x1 * ...	0.000047	(-x1 - 0.35142997)*9.335864e-5* (x1*(-0.3848535...	PySRFunction	
	complexity	loss	equation	score	sympy_format	language	
0	1	95.006485	9.37903	0.000000	9.379030000000000	PySRFunction	(X=>9.3790
1	3	0.182313	x1 * 0.095376894 (-0.5369693 -	3.127988	x1*0.095376894	PySRFunction	(X=>x1*(C

2	5	0.180949	$x_1^* (-0.095127665)$	0.003754	$(-x_1 - 0.5369693)^* (-0.095127665)$	PySRFunction(X=>(-x1 - (-0
3	7	0.027002	$((x_2^* 1.305679e-5) + 0.08985588)^* x_1$	0.951147	$x_1^*(x_2^*1.305679e-5 + 0.08985588)$	PySRFunction(X=>x1*(x2* +
4	9	0.025721	$((x_2^* 1.332342e-5) + 0.08998075)^* (x_1 - 0.50...$	0.024307	$(x_1 - 1^*0.5065427)^* (x_2^*1.332342e-5 + 0.08998075)$	PySRFunction(X=>(x1 - 1 (;
5	11	0.025585	$((1.0000131^* (x_2 + 0.0900638)) - x_2)) + ...$	0.002658	$x_1^*(-x_2 + 1.0000131^*(x_2 + 0.0900638)) - 0.0514...$	PySRFunction(1.0000131
6	13	0.025160	$(x_2^* ((x_0^* -4.5225705e-5) + 0.0025148662)) + ...$	0.008359	$x_2^*(0.0025148662 + x_0^*(-4.5225705e-5)) + (x_1 - ...$	PySRFunction(X=>x2*(0 + ;
7	15	0.025159	$((x_2^* ((x_0^* -4.519881e-5) + 0.0025130874)) - ...$	0.000022	$-x_1 + x_1^*1.1099097 + x_2^*(0.0025130874 + x_0^*(-4...$	PySRFunction(X=>-x1 + x
8	17	0.024587	$((0.043033525 - x_2) - x_2) + (x_2 + (x_2^* 1.000...$	0.011503	$(x_1^*2.0877771 - 1^*1.1884836)^*(-x_2 - x_2 + x_2 + ...$	PySRFunction(X=>(x1 1^*1.
9	21	0.024574	$(((((x_1 - (x_2^* 0.1797244))^* -1.180443e-8)^* ...$	0.000132	$x_1^* (x_2^*1.2642632e-5 + (x_1 - 0.1797244^*x_2)^* (-1....$	PySRFunction(X=>x1*(x2*
10	23	0.024527	$(x_1^* (((x_2 + (((x_1 - (x_2 + -1.2185787))^* (x_1...$	0.000964	$x_1^*((x_2 + (x_1 - (x_2 - 1.2185787))^*(x_1 - 0.1661...$	PySRFunction(X=>x1*((x2
11	25	0.024527	$(x_1^* (((x_2 + ((x_1 - ((x_2 - 0.43084326) + -1.2...$	0.000001	$x_1^*((x_2 + (x_1 - (x_2 - 1.2185787 - 1^*0.43084326...$	PySRFunction(X=>x1*((x2
12	27	0.024520	$(x_1^* (((x_2 + (((x_1 - (((x_2 - x_1) - 0.19967812...$	0.000135	$x_1^*((x_2 + (x_1 - (x_1 - 0.15537381^*x_2)^*(-x_1 + x_2...$	PySRFunction(X=>x1*((x2

- outputs/20250408_024553_nnhAq2/hall_of_fame.csv

```
In [11]: # Stage 3: Compare with LSTM or GRU (placeholder for future development)
print("This is where you'd insert an LSTM/GRU model to compare forecasting results.")
```

This is where you'd insert an LSTM/GRU model to compare forecasting results.

```
In [12]: # Stage 4: Evaluation metrics
from sklearn.metrics import mean_squared_error, r2_score

for name, y_true, y_pred in zip(["S", "I", "R"], [dS_dt, dI_dt, dR_dt],
                                [model_dS.predict(X_dyn), model_dI.predict(X_dyn), model_dR.predict(X_dyn)]):
    print(f"{name} RMSE:", np.sqrt(mean_squared_error(y_true, y_pred)))
    print(f"{name} R²:", r2_score(y_true, y_pred))
```

```
S RMSE: 0.24275533041347475
S R²: 0.9996222289086407
I RMSE: 0.4858872768364714
I R²: 0.9964273281721756
R RMSE: 0.16432398896437092
R R²: 0.9997157839642422
```

```
In [13]: # Stage 5: Mathematical background display

from IPython.display import Markdown
Markdown(r'''
### SIR Model Equations

The standard SIR model is a system of differential equations:

$$
\begin{aligned}
\frac{dS}{dt} &= -\beta S I \\
\frac{dI}{dt} &= \beta S I - \gamma I \\
\frac{dR}{dt} &= \gamma I
\end{aligned}
$$

Where:

-  $(S)$ : Susceptible population
-  $(I)$ : Infected population
-  $(R)$ : Recovered population
-  $(\beta)$ : Transmission rate
-  $(\gamma)$ : Recovery rate
''')
```

Out[13]:

SIR Model Equations

The standard SIR model is a system of differential equations:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dI}{dt} &= \beta SI - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Where:

- (S) : Susceptible population
- (I) : Infected population
- (R) : Recovered population
- (β) : Transmission rate
- (γ) : Recovery rate

In []: